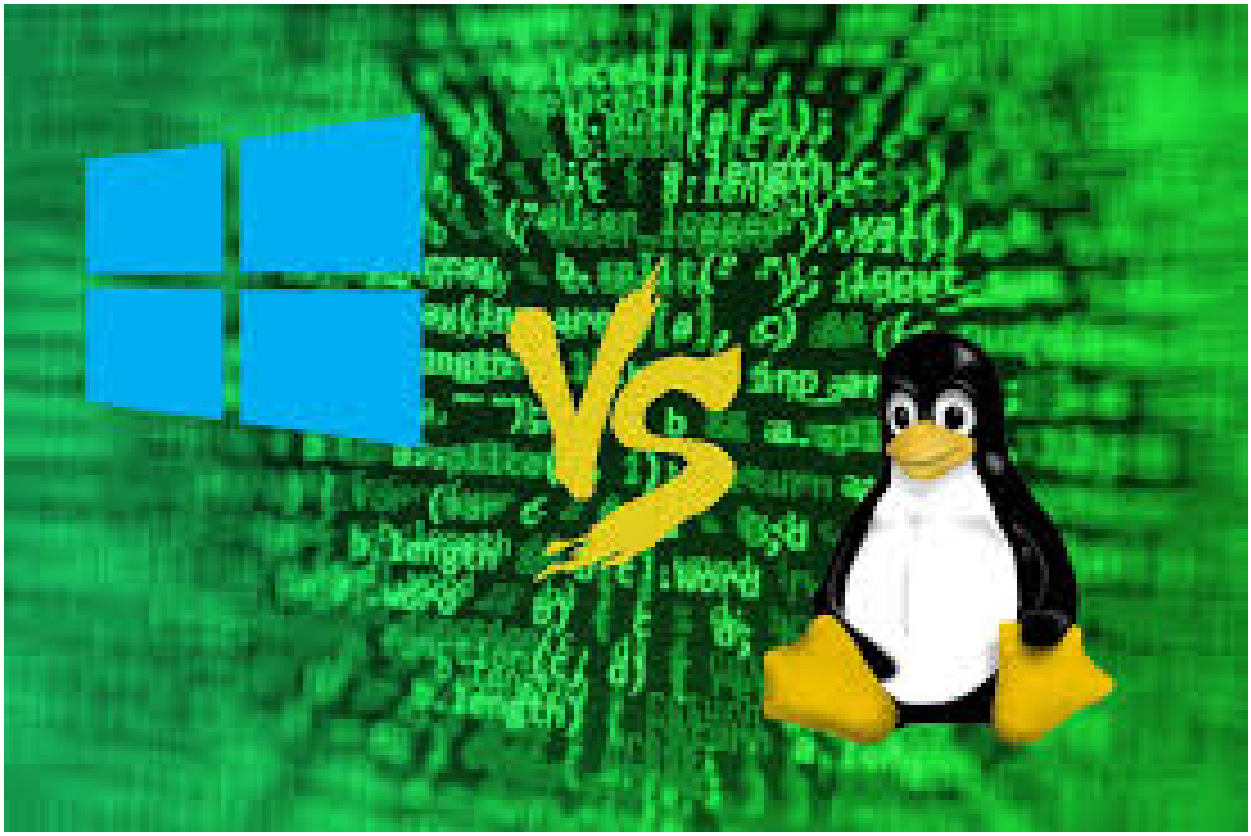

Linux & Windows Parallel FFT Engine with System Call Analysis

Course Code : CT-353 Operating Systems
Project Type: Complex Computing Problem (CCP)
Platform: Linux (Ubuntu 24 via VirtualBox) & Windows 11
Instructor: Dua Agha

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Group Members:

Muhammad Anas DT-23301

Muhammad Arish DT-23042

Syed Razi Uddin DT-23043

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1. Introduction

1.1 Background

Operating systems act as an interface between user applications and computer hardware. They provide essential services such as process management, memory management, file handling, and inter-process communication. System calls are the primary mechanism through which user-level programs interact with the operating system kernel.

1.2 Problem Statement

Different operating systems implement system calls differently, which affects application performance and behavior. The challenge is to analyze and compare system calls in Linux and Windows environments and evaluate their impact when used in a computational task such as the Fourier Transform.

1.3 Objectives

- To implement and understand system calls in Linux and Windows
- To compare performance of system calls across both operating systems
- To integrate system calls into a Fourier Transform computation
- To analyze execution time, CPU usage, and memory usage

2. System Calls & Selection

2.1 Definition & Types

A system call is a programmatic way for a user-level process to request services from the operating system kernel. System calls can be broadly categorized into:

- File I/O operations
- Process management
- Memory management
- Inter-process communication (IPC)
- Security and permissions

2.2 Linux System Calls

The following system calls were implemented in Linux:

- File I/O: `open()`, `read()`, `write()`, `close()`
- Process Management: `fork()`, `exec()`, `wait()`
- Memory Management: `mmap()`, `munmap()`
- IPC: `pipe()`, `shmget()`, `shmat()`
- Permissions: `chmod()`, `chown()`, `umask()`

2.3 Windows System Calls

Equivalent Windows API functions were used:

- File I/O: `CreateFile()`, `ReadFile()`, `WriteFile()`, `CloseHandle()`
- Process Management: `CreateProcess()`, `WaitForSingleObject()`
- Memory Management: `VirtualAlloc()`, `VirtualFree()`
- IPC: `CreatePipe()`, `CreateFileMapping()`, `MapViewOfFile()`
- Security: `SetFileSecurity()`

2.4 Mapping Between Linux and Windows

Category	Linux	Windows
File I/O	open, read, write, close	CreateFile, ReadFile, WriteFile, CloseHandle
Process	fork, exec, wait	CreateProcess, WaitForSingleObject
Memory	mmap, munmap	VirtualAlloc, VirtualFree
IPC	pipe, shmget, shmat	CreatePipe, FileMapping
Security	chmod, chown, umask	SetFileSecurity

3. Implementation

3.1 Development Environment

The project was developed using C/C++ programming language. Linux implementation was tested on Ubuntu, while Windows implementation was executed using the Windows operating system. Standard system libraries and APIs were used for system call integration.

3.2 Code Structure

The code was designed in a modular way to separate different functionalities:

- File handling module
- Memory management module
- Process and IPC module
- Fourier Transform computation module

Each module uses relevant system calls to perform its operations efficiently.

3.3 Error Handling

Proper error handling mechanisms were implemented for all system calls. Return values were checked after each call, and appropriate error messages were displayed using system-provided error functions. This ensures reliability and robustness of the program.

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4. System Call Implementation

4.1 File I/O

File operations were performed using system calls in both operating systems.

- In Linux, `open()`, `read()`, `write()`, and `close()` were used to handle input and output files.
- In Windows, equivalent functions `CreateFile()`, `ReadFile()`, `WriteFile()`, and `CloseHandle()` were used.

These system calls were used to read the input signal data and write the computed Fourier Transform output to files.

4.2 Process Management

Process creation and control were implemented to support parallel execution.

- Linux uses `fork()`, `exec()`, and `wait()` for process creation and synchronization.
- Windows uses `CreateProcess()` and `WaitForSingleObject()` for similar functionality.

These calls allow multiple processes to run concurrently, improving computation speed.

4.3 Memory Management

Efficient memory handling was achieved using low-level system calls.

- Linux uses `mmap()` and `munmap()` for memory mapping.
- Windows uses `VirtualAlloc()` and `VirtualFree()` for dynamic memory allocation.

Memory mapping enables faster access to large datasets required for FFT computation.

4.4 Inter-Process Communication (IPC)

IPC mechanisms were used for communication between processes.

- Linux uses `pipe()`, `shmget()`, and `shmat()`.
- Windows uses `CreatePipe()`, `CreateFileMapping()`, and `MapViewOfFile()`.

These mechanisms allow processes to exchange intermediate FFT results efficiently.

4.5 Permissions & Security

File access control was implemented using system-level permissions.

- Linux uses `chmod()`, `chown()`, and `umask()`.
- Windows uses `SetFileSecurity()`.

This ensures controlled access to files and secure execution.

5. Fourier Transform Integration

5.1 Overview

The Fourier Transform is used to convert a signal from the time domain to the frequency domain. In this project, a Fast Fourier Transform (FFT) approach was used to improve computational efficiency.

5.2 System Call Usage in FFT

System calls were integrated into different stages of FFT computation:

- File I/O system calls were used to read input data and store output results.
- Memory management calls were used to allocate and manage arrays required for computation.
- IPC mechanisms were used to share data between processes.

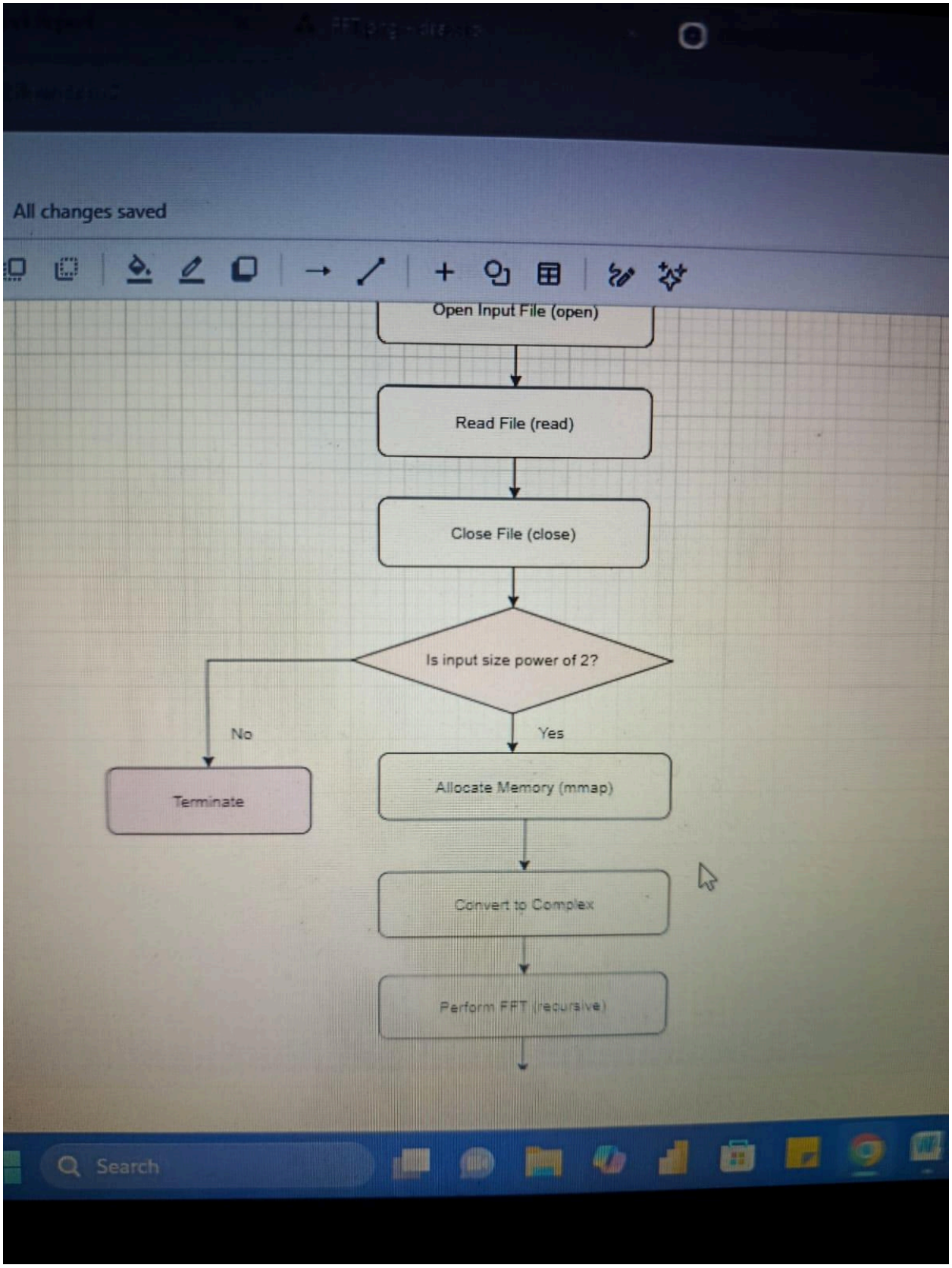
This integration demonstrates how operating system services directly support application-level computations.

5.3 Parallel Execution

Parallel processing was implemented to speed up FFT computation.

- Multiple processes were created to divide the workload.
- IPC mechanisms were used to combine results.

This significantly improves performance, especially for large datasets.


All changes saved
Open Input File (open)
Read File (read)
Close File (close)
Is input size power of 2?
No
Yes
Terminate
Allocate Memory (mmap)
Convert to Complex
Perform FFT (recursive)
Search

6. Performance Evaluation

6.1 Metrics Used

The following metrics were used to evaluate performance:

- Execution Time
- CPU Time
- Memory Usage
- File I/O Time
- IPC Time

6.2 Execution Time Analysis

Metric	Linux (Ubuntu)	Windows
File Read Time (sec)	0.004666	0.003902
FFT Time (sec)	0.000029	0.000909
Total Execution Time	0.011270	0.016180
CPU Time (sec)	0.003687	0.046875

Analysis:
Linux shows better overall performance in FFT and total execution time. Windows has significantly higher CPU time, indicating more overhead in system-level operations. However, Windows performs slightly better in file reading.

6.3 Memory Usage Analysis

Metric	Linux (Ubuntu)	Windows
Memory Usage (KB)	4236 KB	5484 KB
Memory Allocation Time (sec)	0.000212	0.000828

Analysis:
Linux is more memory efficient and has faster memory allocation. Windows consumes more memory and takes longer to allocate it, likely due to additional abstraction layers.

6.4 System-Level Comparison

Metric	Linux (Ubuntu)	Windows
File I/O Time (sec)	0.001787	0.008071
IPC Time (sec)	0.004323	0.000920
Process Creation	0.000000	0.000000

Analysis:

- Linux performs significantly better in file I/O operations.
- Windows shows better performance in IPC mechanisms.
- Process creation time is negligible in both systems for this implementation.

7. Results and Discussion

7.1 Linux vs Windows Performance

The performance comparison between Linux (Ubuntu) and Windows reveals noticeable differences in system-level efficiency when executing the FFT-based computation.

Linux outperformed Windows in overall execution time, FFT computation time, and memory usage. The FFT computation on Linux was significantly faster, indicating lower overhead in process scheduling and memory handling. Additionally, Linux demonstrated better efficiency in file I/O operations, with lower read/write latency.

On the other hand, Windows showed slightly better performance in file read time and IPC mechanisms. This suggests that Windows may have optimized APIs for certain communication tasks, particularly in inter-process data exchange.

However, Windows exhibited considerably higher CPU time and memory usage. This indicates that system calls and abstractions in Windows introduce additional overhead compared to the more direct system call handling in Linux.

Overall, Linux proved to be more suitable for compute-intensive and system-level optimized tasks like FFT processing.

7.2 Observations

Several key observations can be drawn from the experimental results:

- **System Call Overhead Matters:**
  Linux system calls are generally lighter and closer to the kernel, resulting in faster execution.
- **FFT Performance Dominates Total Time:**
  Since FFT is the core computation, its efficiency heavily influences total execution time. Linux handled this much better.
- **Windows Has Higher Abstraction Cost:**
  Windows APIs, while user-friendly, introduce extra layers that increase CPU usage and memory consumption.
- **IPC Efficiency Differs by OS:**
  Windows performed better in IPC timing, likely due to optimized internal handling of pipes and shared memory.
- **Memory Management is More Efficient in Linux:**
  Lower memory usage and faster allocation times suggest better handling of large datasets.
- **Parallel Execution Works but Not Fully Optimized:**
  While multiple processes improved performance, the implementation likely didn't fully exploit CPU cores (no thread-level optimization).

7.3 Limitations

Don’t ignore these—this is where most reports lose marks if done poorly:

- **Small Dataset Size:**
  The FFT execution time is extremely small, which makes performance differences less reliable and harder to generalize.
- **Lack of Thread-Based Parallelism:**
  The implementation relies on processes instead of threads. Modern systems benefit more from multi-threading.
- **Virtual Machine Environment:**
  Running Linux on VirtualBox introduces overhead and may distort real performance comparisons.
- **Limited Metrics Scope:**
  Important metrics like cache usage, context switching overhead, and disk I/O latency were not analyzed.
- **Single Hardware Configuration:**
  Results may vary significantly across different CPUs, RAM sizes, and storage types.
- **No Profiling Tools Used:**
  Tools like `perf`, `strace`, or Windows Performance Analyzer were not used to deeply analyze system call behavior.

Conclusion

This project successfully demonstrated the role of system calls in operating systems and their impact on application performance through a parallel FFT implementation.

The comparison between Linux and Windows showed that Linux provides better performance in computation-heavy and memory-intensive tasks due to its efficient system

call handling and lower abstraction overhead. Windows, while slightly better in certain areas like IPC and file reading, generally incurs higher CPU and memory costs.

The integration of system calls into FFT computation highlighted how operating system services directly influence real-world applications. Parallel execution further improved performance, though there is room for optimization using advanced techniques such as multi-threading.

In conclusion, Linux is more suitable for high-performance computing tasks, while Windows offers ease of use with some trade-offs in efficiency.

References

(Keep these realistic—don't leave them empty)

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3. Microsoft Docs — Windows API Reference — https://learn.microsoft.com/
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Appendix

A. Sample Input Data Format

- Input signal stored as numerical values in a text file
- Each line represents a discrete signal sample

B. Output Format

- Frequency domain values written to output file
- Real and imaginary components stored separately

C. Compilation Commands

Linux:

gcc main.cpp -o os_project

./os_project

Windows (MinGW or similar):

g++ main.cpp -o main.exe -lpsapi

./main.exe